Jahangirnagar University
জাহাঙ্গীরনগর বিশ্ববিদ্যালয়

Department of Computer Science and Engineering

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2nd Year 1st Semester (Hons)

Course Title: Digital Logic Design Laboratory

Course Code: CSE-254

Experiment No: 05

Experiment Title: Design and implement 3:8 decoder using 2:4 decoder

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26.8.25

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Experiment No: 5

Experiment Name:

Design and implement 3:8 decoder using 2:4 decoder.

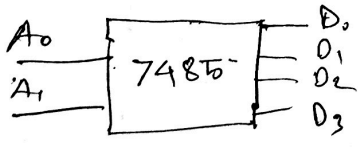
Objective:

- To understand the function and operation of a 2:4 and a 3:8 decoder.
- To design the logic circuit for a 2:4 decoder

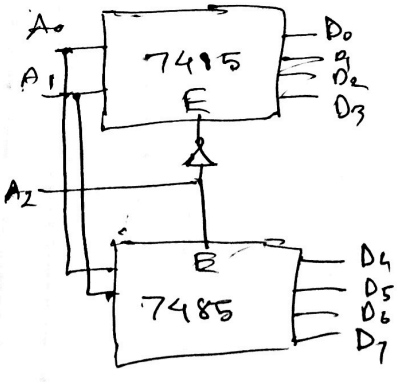
Required components:

Digital trainer board, ICs (7404, 7408, 7432), connecting wires,

Block Diagram:

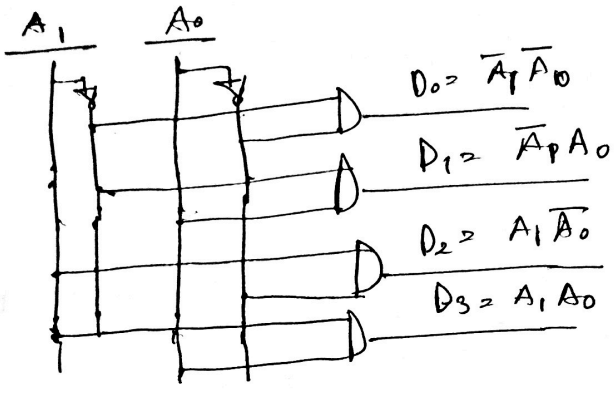


2:4 Decoder

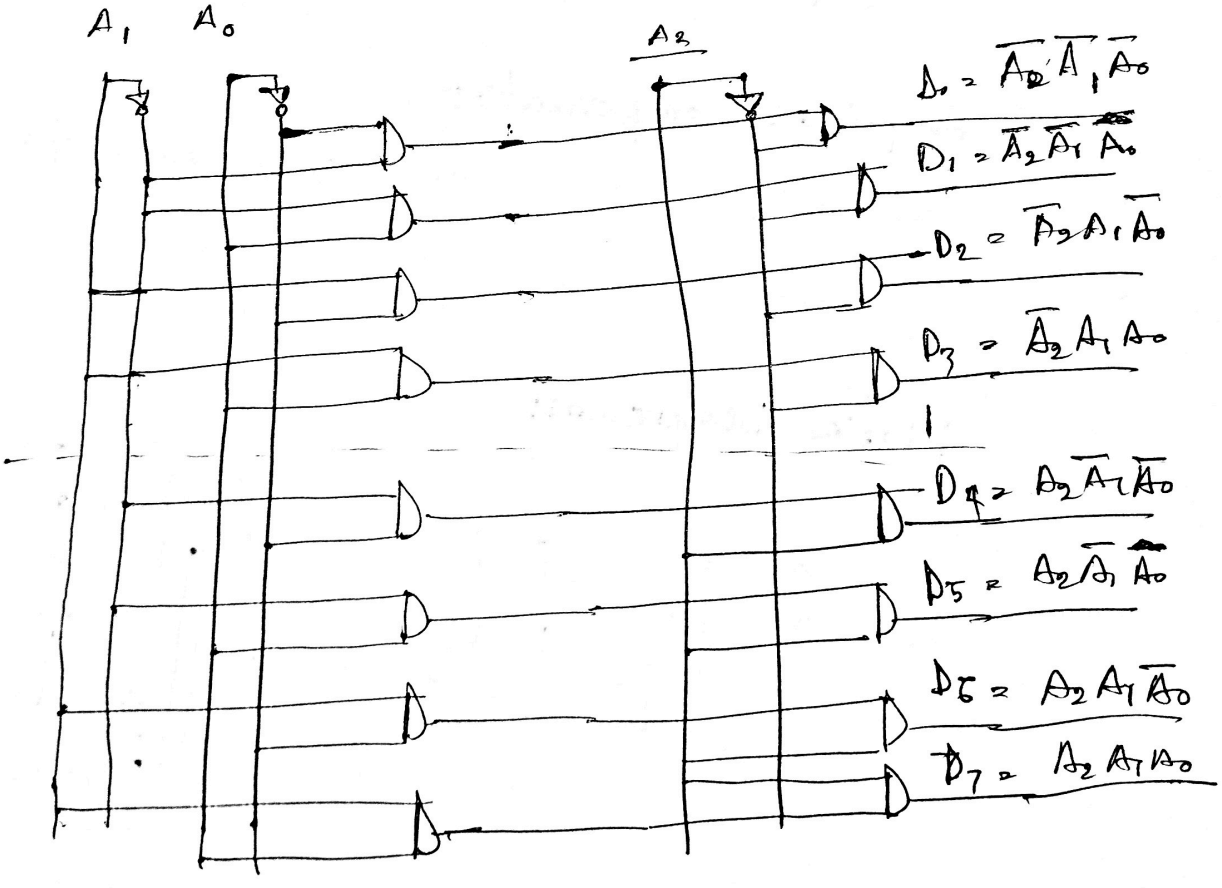


3:8 decoder

Circuit Diagram:



2:4 decoder circuit



3:8 Decoder using 2 2:4 decoders

Truth Tables:

A_1	A_0	D_0	D_1	D_2	D_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

A_2	A_1	A_0	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

Procedure:

- Place the ICs on the breadboard
- Connect the wires with corresponding gates
- Check the LED outputs on the transparent board.

Result:

All LED worked successfully. The truth table is totally satisfied. So, all the signals were decoded correctly

Discussion:

The experiment ensures the decoding of digital signals using logic gates in which for one combination there's only one true value.